BONUS · NEUROKIDS

Space Activities

Explore the universe with fun puzzles, facts, and hands-on space adventures

Table of Contents

Welcome to Space! 3

Planet Fact Cards 4

Solar System Ordering 6

Space Maze 7

Space Word Search 8

Constellation Connect-the-Dots 9

Design Your Own Rocket 12

Alien Creature Design 13

Moon Phases Matching 14

Space Counting & Math 15

Astronaut Checklist Role-Play 16

Space Journal Writing Prompt 17

Star Pattern Completion 18

Space Bingo Card 19

1

Welcome to Space!

Amazing facts and cosmic wonders await you

> *"Space is for everybody. It's not just for a few people in science or math, or for a select group of astronauts. That's our new frontier out there." — Christa McAuliffe*

Get ready for an incredible journey through the cosmos! In this activity book, you will discover the planets, explore constellations, solve puzzles, and even design your own rocket. Here are some mind-blowing facts to kick off your space adventure:

Amazing Space Fact #1

The Sun is so big that about 1.3 million Earths could fit inside it. Even though it looks small in our sky, the Sun is actually a medium-sized star that measures 864,000 miles across!

Amazing Space Fact #2

A day on Venus is longer than a year on Venus. It takes Venus 243 Earth days to spin once on its axis, but only 225 Earth days to orbit all the way around the Sun.

Amazing Space Fact #3

There are more stars in the universe than grains of sand on all of Earth's beaches. Scientists estimate there are around 200 billion trillion stars out there—that is a 2 followed by 23 zeros!

Amazing Space Fact #4

Neutron stars are so dense that a teaspoon of their material would weigh about 6 billion tons on Earth. That is roughly the weight of every car, truck, and bus in the world combined!

Amazing Space Fact #5

The footprints left on the Moon by the Apollo astronauts will stay there for millions of years. The Moon has no wind or rain to wash them away, so those footprints from 1969 are still exactly as they were left.

Amazing Space Fact #6

Saturn is so light that if you could find a bathtub big enough, it would float! Saturn's density is less than that of water because the planet is made mostly of hydrogen and helium gas.

2

Planet Fact Cards

Learn about each planet in our Solar System

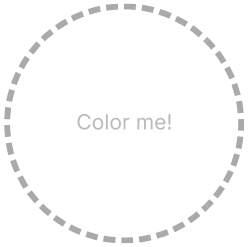
Mercury

Position from Sun: 1st

Size: Mercury is about the size of Earth's Moon.

Fun Fact: Mercury has no atmosphere, so its surface temperature swings from 800 °F during the day to −290 °F at night.

Color: dark gray



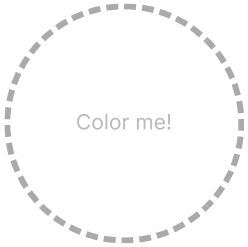
Venus

Position from Sun: 2nd

Size: Venus is nearly the same size as Earth (sometimes called Earth's "twin").

Fun Fact: Venus spins backwards compared to most planets, so the Sun rises in the west and sets in the east.

Color: pale yellow-white



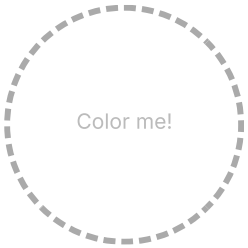
Earth

Position from Sun: 3rd

Size: Earth's diameter is about 7,918 miles—just right for life!

Fun Fact: Earth is the only known planet where liquid water exists on the surface, covering about 71% of it.

Color: blue and green



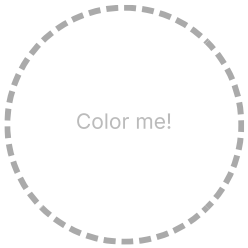
Mars

Position from Sun: 4th

Size: Mars is about half the size of Earth.

Fun Fact: Mars has the tallest volcano in the Solar System—Olympus Mons, which is nearly three times the height of Mount Everest.

Color: reddish-orange



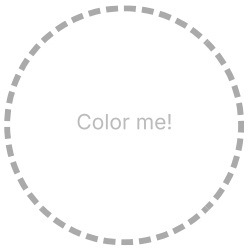
Jupiter

Position from Sun: 5th

Size: Jupiter is the largest planet—more than 1,300 Earths could fit inside it.

Fun Fact: Jupiter's Great Red Spot is a storm that has been raging for at least 350 years and is bigger than Earth.

Color: orange-tan with brown bands



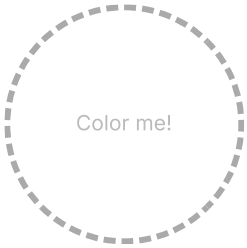
Saturn

Position from Sun: 6th

Size: Saturn is the second largest planet, about 9.5 times wider than Earth.

Fun Fact: Saturn's famous rings are made mostly of chunks of ice and rock, some as tiny as grains of sand and some as big as houses.

Color: golden-yellow



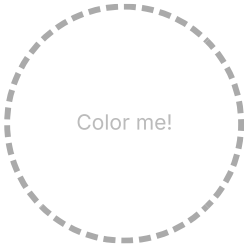
Uranus

Position from Sun: 7th

Size: Uranus is about 4 times wider than Earth.

Fun Fact: Uranus rotates on its side, rolling around the Sun like a ball. Scientists think a massive collision long ago tipped it over.

Color: pale blue-green



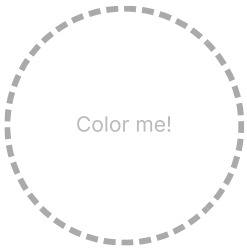
Neptune

Position from Sun: 8th

Size: Neptune is about 3.9 times wider than Earth.

Fun Fact: Neptune has the strongest winds in the Solar System, reaching speeds of 1,200 miles per hour—faster than the speed of sound on Earth.

Color: deep blue



Solar System Ordering

The planets are all mixed up! Can you write them in the correct order, from closest to the Sun to farthest away?

Scrambled Planets:

Saturn | Earth | Neptune | Mercury | Mars | Uranus | Venus | Jupiter

Write them in order (1 = closest to the Sun):

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____

> **Helpful Hint**
>
> Try this memory trick: **M**y **V**ery **E**xcellent **M**other **J**ust **S**erved **U**s **N**achos. The first letter of each word matches the first letter of each planet, in order!

Space Maze

Help the astronaut find the path to the spaceship! Draw a line from START to FINISH. Watch out for dead ends along the way.

START: Astronaut **FINISH: Spaceship**



Tip: Try starting from the FINISH and working backwards if you get stuck. Use a pencil so you can erase wrong turns!

Space Word Search

Find all 10 space words hidden in the grid below. Words can go across (left to right) or down (top to bottom). Circle each word when you find it!

G	A	L	A	X	Y	D	R	W	M
P	F	H	J	V	Q	I	O	B	A
L	W	D	M	O	O	N	C	U	R
A	N	E	B	U	L	A	K	P	S
N	G	O	R	B	I	T	E	F	H
E	Y	C	O	M	E	T	T	J	D
T	K	V	W	H	F	Q	L	B	G
S	O	L	A	R	D	I	N	E	P
W	B	F	J	U	H	S	T	A	R
D	Q	K	P	G	V	C	E	N	L

Find these 10 words:

PLANET STAR MOON ROCKET COMET ORBIT GALAXY NEBULA SOLAR MARS

> **Directions**
>
> 6 words go **across** (left to right): GALAXY, MOON, NEBULA, ORBIT, COMET, SOLAR.
> 4 words go **down** (top to bottom): PLANET, ROCKET, STAR, MARS.

Constellation Connect-the-Dots

Each constellation is a pattern of stars in the night sky. Connect the numbered dots in order (1, 2, 3...) to reveal each constellation shape!

Big Dipper — 7 Stars

The Big Dipper is one of the easiest patterns to spot in the northern sky. It looks like a large ladle or cup with a long handle. It is part of the larger constellation Ursa Major (the Great Bear).

How to connect: Dots 1 through 4 form the rectangular bowl. Dots 4 through 7 form the curved handle.

1 → 2 → 3 → 4 → 1 (bowl), then 4 → 5 → 6 → 7 (handle)

Draw the Big Dipper here — place 7 numbered dots and connect them

Orion (The Hunter) — 7 Stars

Orion is visible from almost everywhere on Earth. Look for the three bright stars in a row that form Orion's Belt. Betelgeuse (top left) glows red, and Rigel (bottom right) shines blue-white.

How to connect: Dots 1–2 are the shoulders (top). Dots 3–4–5 are the belt (middle row). Dots 6–7 are the feet (bottom).

1 → 3 → 6 (left side), 2 → 5 → 7 (right side), 3 → 4 → 5 (belt)

Draw Orion here — place 7 numbered dots and connect them

Cassiopeia — 5 Stars

Cassiopeia looks like a big letter W (or an M, depending on the time of night). It is named after a queen from Greek mythology and is visible year-round in the northern hemisphere.

How to connect: Connect the dots in order 1 → 2 → 3 → 4 → 5 to form a zigzag W shape.

1 (upper left) → 2 (lower) → 3 (upper middle) → 4 (lower) → 5 (upper right)

Draw Cassiopeia here — place 5 numbered dots in a W shape and connect them

Leo (The Lion) — 9 Stars

Leo is best seen in spring. The brightest star, Regulus, marks the lion's heart. The front part of Leo looks like a backwards question mark (called the "Sickle"), and the back forms a triangle.

How to connect: Dots 1 through 6 form the Sickle (the lion's head and mane) in a curve. Dots 6 → 7 → 8 → 9 → 6 form the triangle (the lion's body and tail).

Sickle: 1 → 2 → 3 → 4 → 5 → 6. Triangle: 6 → 7 → 8 → 9 → 6.

Draw Leo here — place 9 numbered dots and connect them

Scorpius (The Scorpion) — 8 Stars

Scorpius is a summer constellation in the southern sky. Its brightest star, Antares, glows red and represents the scorpion's heart. The long chain of stars curves into a hook shape that forms the scorpion's tail and stinger.

How to connect: Dots 1–2–3 form the head (top). Dot 4 is the heart (Antares). Dots 4 → 5 → 6 → 7 → 8 curve downward and hook back up to form the tail and stinger.

1 → 2 → 3 (head), then 2 → 4 → 5 → 6 → 7 → 8 (body and tail)

Draw Scorpius here — place 8 numbered dots and connect them

Stargazing Tip

To see constellations best, go outside on a clear night away from bright city lights. Give your eyes about 15 minutes to adjust to the dark. Lie on a blanket and look straight up. A star chart or stargazing app can help you find each constellation!

Design Your Own Rocket

You are a rocket engineer! Design your very own spacecraft below. Draw each part in the correct section, then fill in the details.

NOSE CONE (the very top — the pointed tip that cuts through the air)

BODY (the middle — this is where the crew sits and cargo is stored)

FINS (the bottom — these keep the rocket flying straight)

Name your rocket:

Special feature (what makes your rocket unique?):

Top speed:

Mission destination (where is it going?):

Alien Creature Design

Imagine you discovered a new alien species on a faraway planet! Draw your alien in the space below, then fill in the details about your creature.

Draw your alien here:

How many eyes does it have?

How many arms does it have?

What color is it?

What does it eat?

What planet is it from?

Give it a name:

Does it have any special powers? Describe them:

Moon Phases Matching

The Moon changes shape throughout the month as it orbits Earth. Match each moon phase name on the left with the correct description on the right. Write the letter of the description on the line next to each phase.

Moon Phase Names:

1. **New Moon** Match: _____
2. **Waxing Crescent** Match: _____
3. **First Quarter** Match: _____
4. **Waxing Gibbous** Match: _____
5. **Full Moon** Match: _____
6. **Waning Gibbous** Match: _____
7. **Third Quarter** Match: _____
8. **Waning Crescent** Match: _____

Descriptions (scrambled):

LETTER	DESCRIPTION
A	The entire face of the Moon is brightly lit up.
B	A thin sliver of light appears on the right side, growing larger.
C	The left half of the Moon is lit; the right half is dark.
D	The Moon is not visible from Earth because the lit side faces away.
E	More than half the Moon is lit and still growing toward full.
F	Only a thin sliver of light remains on the left side before it disappears.
G	The right half of the Moon is lit; the left half is dark.
H	The light begins to shrink from the right side after the full moon.

> **Remember**
>
> "Waxing" means the lit part is **growing**. "Waning" means the lit part is **shrinking**. The cycle from New Moon back to New Moon takes about 29.5 days.

Space Counting & Math

Solve each space-themed word problem. Write your answer on the line. Show your work if you like!

Problem 1: There are 3 rockets on the launch pad. 2 more arrive. How many rockets are there now?

Answer: _____

Problem 2: An astronaut collected 9 moon rocks. She gave 4 to the science lab. How many moon rocks does she still have?

Answer: _____

Problem 3: There are 8 planets in our Solar System. If you have already learned about 5 of them, how many more planets do you still need to learn about?

Answer: _____

Problem 4: A space station has 6 astronauts. A new crew of 3 astronauts arrives on a supply ship. How many astronauts are on the station now?

Answer: _____

Problem 5: Jupiter has 4 very large moons (called the Galilean moons). Saturn also has 1 very large moon (Titan). How many large moons is that in total?

Answer: _____

Problem 6: A rocket blasts off and passes through 3 layers of the atmosphere in the first minute. In the second minute, it passes through 2 more layers. How many layers has it passed through in all?

Answer: _____

Problem 7: There are 10 stars in a constellation. An astronomer can see 7 of them tonight because clouds are hiding the rest. How many stars are hidden by the clouds?

Answer: _____

Problem 8: A satellite orbits Earth 4 times each day. How many times does it orbit Earth in 3 days?

Answer: _____

Astronaut Checklist Role-Play

Imagine you are an astronaut about to go on a real space mission! Follow this checklist step by step. Check off each item as you pretend to do it. You can act out the whole mission at home!

☐ Put on your space suit and helmet. ☐ Check your oxygen tank levels — is it full? Say "Oxygen: check!" ☐ Buckle your five-point seat belt in the command chair. ☐ Run the instrument panel check — flip all the switches to ON. ☐ Radio Mission Control: "This is Astronaut [your name], ready for launch!" ☐ Start the countdown: 10... 9... 8... 7... 6... 5... 4... 3... 2... 1... ☐ BLAST OFF! Feel the rumble as the engines fire. ☐ Watch the sky change from blue to black as you leave the atmosphere. ☐ Unbuckle your belt and float in zero gravity! ☐ Look out the window — can you see Earth from up here? ☐ Eat a pouch of astronaut food (squeeze it into your mouth!). ☐ Collect a moon rock and put it in your sample bag. ☐ Take a photo of the stars for the science team back home. ☐ Radio Mission Control: "Heading home!" ☐ Buckle up for re-entry. Hold on tight! ☐ Splash down safely on Earth. Mission complete!

Real Astronaut Fact

Before every real space mission, astronauts train for years and follow checklists with hundreds of items. Checklists help them remember every important step so nothing is missed. NASA astronauts even have special wrist-mounted checklist booklets they wear during spacewalks!

Space Journal Writing Prompt

Imagine you just boarded a spaceship for an incredible mission. Write about your adventure! Use the guiding questions below to help you get started.

Think About These Questions

Where in space are you going? What does the inside of your spaceship look like? Who is on your crew? What do you see out the window? What do you eat? What is the most exciting thing that happens? Do you meet any aliens? What do you bring back to Earth?

"If I went to space..."

Star Pattern Completion

Look at each pattern carefully. Figure out what comes next and draw or write the correct shape in the empty boxes.

Pattern 1: Star – Moon – Star – Moon – Star – ?



Pattern 2: Star – Star – Moon – Star – Star – Moon – Star – Star – ?



Pattern 3: Sun – Star – Moon – Sun – Star – Moon – Sun – ? – ?



Pattern 4: Star – Moon – Moon – Star – Moon – Moon – Moon – Star – Moon – Moon – Moon – Moon – ?



Hint: Count how many moons appear between each star. Is the number growing?

Pattern 5: 1 Star – 2 Stars – 3 Stars – 5 Stars – 8 Stars – ?



Hint: Add the last two numbers together to find the next one. (1+2=3, 2+3=5, 3+5=8, 5+8=?)

Space Bingo Card

Play Space Bingo with friends or family! Have someone call out items from the list. When you hear one on your card, mark it with an X. Get 5 in a row (across, down, or diagonal) to win!

B	I	N	G	O
Shooting Star	Full Moon	Rocket	Astronaut	Planet
Alien	Comet	Satellite	Telescope	Sun
Galaxy	Black Hole	FREE	Meteor	Space Station
Constellation	Nebula	Asteroid	Mars Rover	Eclipse
Star	Orbit	UFO	Solar System	Ring (Saturn)

How to Play

1. One person is the "caller." The caller reads items from the list in random order.
2. When you hear an item that is on your card, mark it with an X or place a small object on it.
3. The FREE space in the middle is already yours—mark it right away!
4. The first player to mark 5 in a row (horizontal, vertical, or diagonal) shouts "BINGO!" and wins.
5. Want to play again? Print extra copies and keep going!

Next Step

Open the NeuroKids main guide and pick your first weekly activity plan to keep the learning going!